EXPANDED CRYPTO STRATEGY RESEARCH

Signals, lower-turnover momentum, trend following, and BTC-factor reversal

Supplement to the original course project | Historical research through September 30, 2026

Executive summary

2 of 8 frozen evaluations had positive 20 bps net returns during January-September 2026. The three 2024-selected extension champions all had negative net returns: Trend long/cash 14d daily -8.04%, Trend long/short 14d daily -16.53%, CS momentum 14d weekly -23.59%.

Scale reversal 28d/weekly had the highest observed net return, 17.11%, with Sharpe 1.53. This is a preserved reference-model result, not a post-holdout selection.

Fixed BTC-factor reversal returned 0.23% net from only 4 spread entries. Sparse activity does not establish a persistent profitable edge.

1 of 24 holdout signal comparisons passed the within-period Holm 5% threshold. Correlations are diagnostic, not independent strategy profits.

Research extension	What it examines
Signal diagnostics	24 lookback/forward-horizon combinations; daily cross-sectional ranks
Lower-turnover momentum	14/28-day lookbacks; daily/weekly schedules; whole-target no-trade buffer
Time-series trend following	Own-coin trends; separately evaluated long/cash and long/short
BTC-factor reversal	Monthly past-data fit; stationarity/half-life gates; aggregate BTC hedge
Regimes	Volume, volatility, dispersion; descriptive following-day contrasts only

Interpretation

The three tuned champions were selected using 2024 only. Four historical reference selections and one fixed residual model complete the eight-model comparison. The nine-month 2026 period was new to this analysis and was accessed only after the freeze. It is a historical holdout, not prospective live evidence. Earlier periods and the inspected 2025 year are explicitly exploratory or retrospective.

Educational unconstrained backtests, not a recommendation to trade. A fixed survivor universe, idealized execution, and excluded short financing limit generalization. Original artifacts remain unchanged.

02 / Data, chronology, and accounting

Partition	Dates	Use
Warm-up	Oct-Dec 2021	Trailing signals and residual formation only
Development	2022-2023	Design, implementation, regime medians
Selection	2024	Highest net Sharpe at 20 bps within each tuned family
Retrospective	2025	Robustness; already inspected, not untouched
Historical holdout	Jan 1-Sep 30, 2026	273 days / 6,552 elapsed hours; no reselection

Point-in-time safeguards

Freeze receipt pins candidate IDs, analysis code, model rules, development-only thresholds, and selections. Higher 2024 net daily Sharpe wins; ties use lower full-L1 turnover, then canonical identifier. Defined negative winners remain research controls. Monthly residual refits are predeclared past-data operations, not tuning on holdout outcomes.

Audit event	UTC timestamp
Selection / analysis freeze	2026-10-06 16:33:04.126074+00:00
First recorded holdout access	2026-10-06 16:33:34.611398+00:00

Return and cost definitions

The existing tested simulator is reused without changing its public API. Signals on completed UTC endpoints earn subsequent intervals with lag one hour; lag two is a stress. Net return equals weighted asset return minus cost, where cost = (bps / 10,000) x sum of absolute executed weight changes. Initial cash entry and terminal liquidation are charged. BTC hedges are aggregated before turnover, so offsetting virtual hedge legs are not double-charged.

Daily compounded returns use 365-day annualization and zero risk-free rate. Total return is the actual nine-month compounded return; CAGR is an annualized transformation, not a twelve-month observation. Daily OLS against BTC includes an intercept; alpha and its HAC(7) 95% interval are annualized, beta is dimensionless. Maximum drawdown uses observed-endpoint equity including initial unit capital.

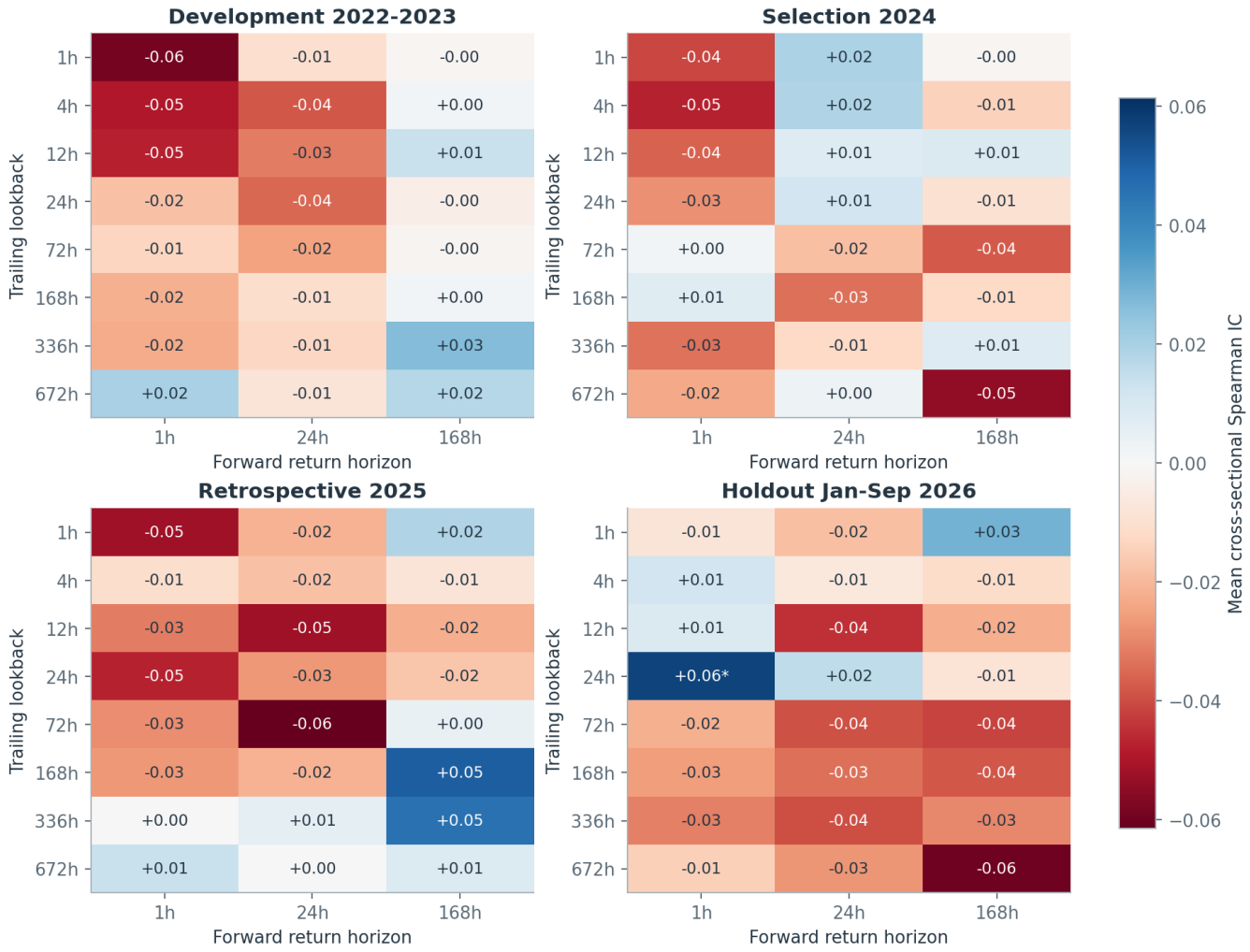
Data quality

Full cache: 480 checksum-verified monthly archives, 350,584 observed of 350,592 expected asset-hour rows; 0 duplicates. All 8 missing asset-hours belong to the documented 2023 halt; the 2026 holdout has no missing hours. Status: pass with documented exceptions, not an exception-free clean pass.

Hourly Binance public spot archives are checksum-verified, cached, and normalized. The original documented March 24, 2023 suspension is preserved: hold invested positions across the 11:00-15:00 UTC observed interval; no fabricated interim prices, executions, or preemptive cash-out. Unknown gaps, missing archives, or failed checksums stop evaluation. Dataset receipts contain metadata repairs and all documented exceptions. No intrahalt drawdown can be reconstructed.

Course approximation: target-change turnover omits ordinary weight drift. Spot short positions are theoretical; borrowing, funding, financing, collateral/margin, capacity, and reliable close fills are not modeled. USDT is the quote numeraire.

03 / Do ranks predict later relative returns?



Daily signals at 00:00 UTC compare cross-sectional past-return ranks with later return ranks across the same eight coins. Positive mean Spearman IC supports momentum; negative supports reversal. Constant or missing cross-sections, suspended endpoints, and forward labels crossing a partition boundary are excluded. The complete daily and summary records remain inspectable.

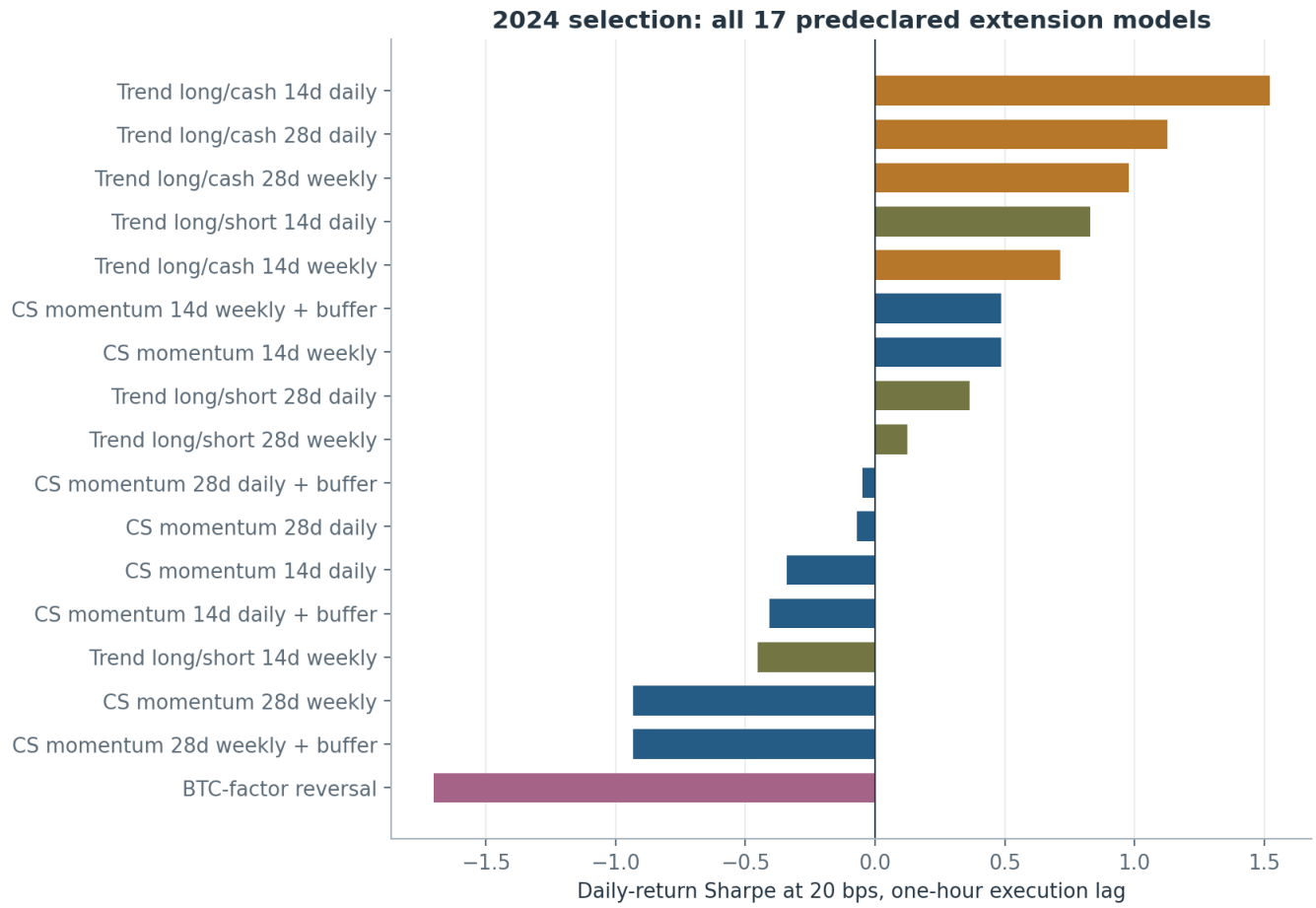
Uncertainty and multiplicity

Mean-IC 95% intervals use 4,000 seeded, 28-day block-bootstrap replicates. Resampling retains within-block serial structure; intervals and p-values are nominal under this declared block scheme, not guaranteed for all dependence. A star marks a Holm-adjusted p-value below 5% across the 24 comparisons within that period. The color scale is shared across panels. Unstarred cells are not established effects; statistically detectable IC is not an execution-cost-adjusted trading profit.

The holdout passing comparison is 24h lookback / 1h forward: mean IC 0.0568, Holm p 0.0480. The adjusted p-value is borderline at 5% and has finite 4,000-replicate Monte Carlo precision. This nominal-bootstrap result is not a cost-aware trading strategy.

All confidence intervals, counts, raw and corrected p-values are retained in outputs/ic_summary.csv and displayed in the notebook. This diagnostic does not select a new 2026 trading horizon.

04 / Historical screening and frozen champions



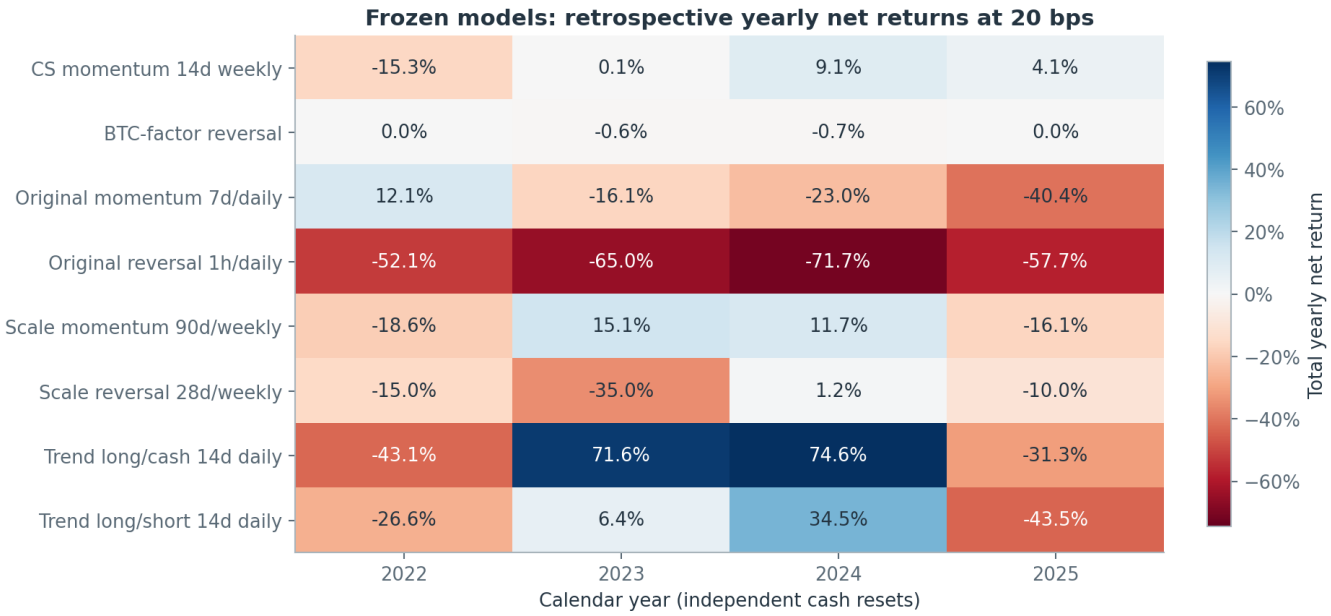
Grid and behavior

Eight cross-sectional configurations combine 14/28-day lookbacks, daily/Monday-weekly schedules, and buffer off/on. The buffer skips the entire decision only when proposed full-L1 target change is below 0.25; otherwise it executes the complete target. Eight time-series configurations use the same lookbacks/schedules for long/cash and long/short. Each coin receives 1/8 when active; inactive allocations are not redistributed.

The fixed residual model is not selected from a hyperparameter grid. Four unbuffered cross-sectional configurations were already examined in the 156-configuration time-scale scan: matching them here is a control comparison, not a new discovery. Historical and failed candidates remain in the CSV.

2024-selected champion	Net total return	Sharpe	Turnover
CS momentum 14d weekly	9.10%	0.49	52.6
Trend long/cash 14d daily	74.60%	1.52	43.5
Trend long/short 14d daily	34.48%	0.83	87.0

05 / Earlier years are retrospective evidence



Identical choices, matched accounting

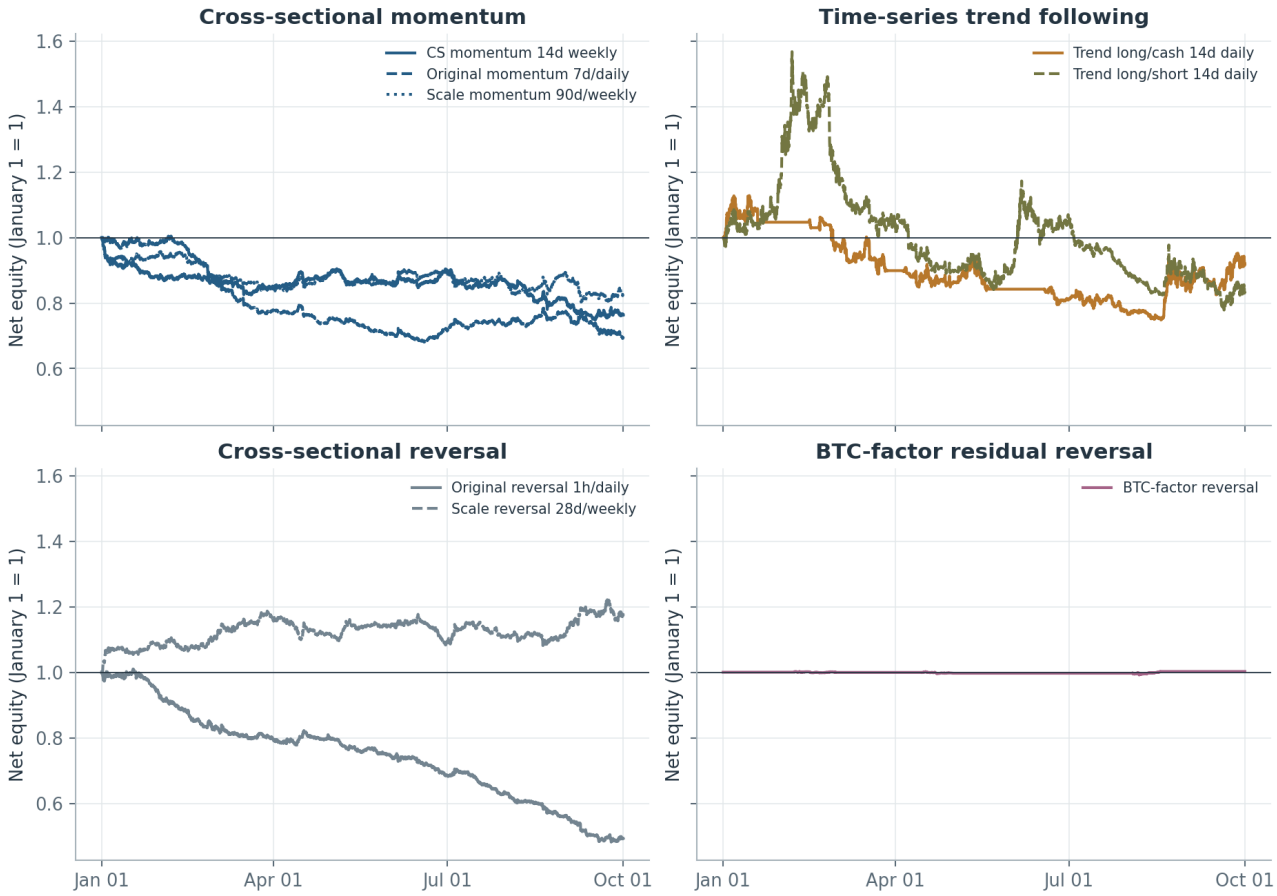
Each row is a frozen champion, fixed residual model, or original/time-scale reference selection. Annual calculations start from cash and liquidate at each year end, charging both transitions. These calculations use the same selected strategy across years, not a newly chosen winner for each year.

2022-2023 is development evidence; 2024 determines the three tuned champions; 2025 is retrospective robustness because the earlier project and time-scale scan already exposed that year. In particular, favorable 2025 returns here cannot be called untouched validation or used to justify reselection after the fact.

What this view can establish

Year-by-year differences expose path sensitivity and how one selected formulation behaved in different observed markets. They do not establish a market-regime mechanism, prospective profitability, or a representative point-in-time investment universe. The full annual-metric CSV includes all required metrics and independent cash-reset flags.

06 / Nine-month historical holdout results

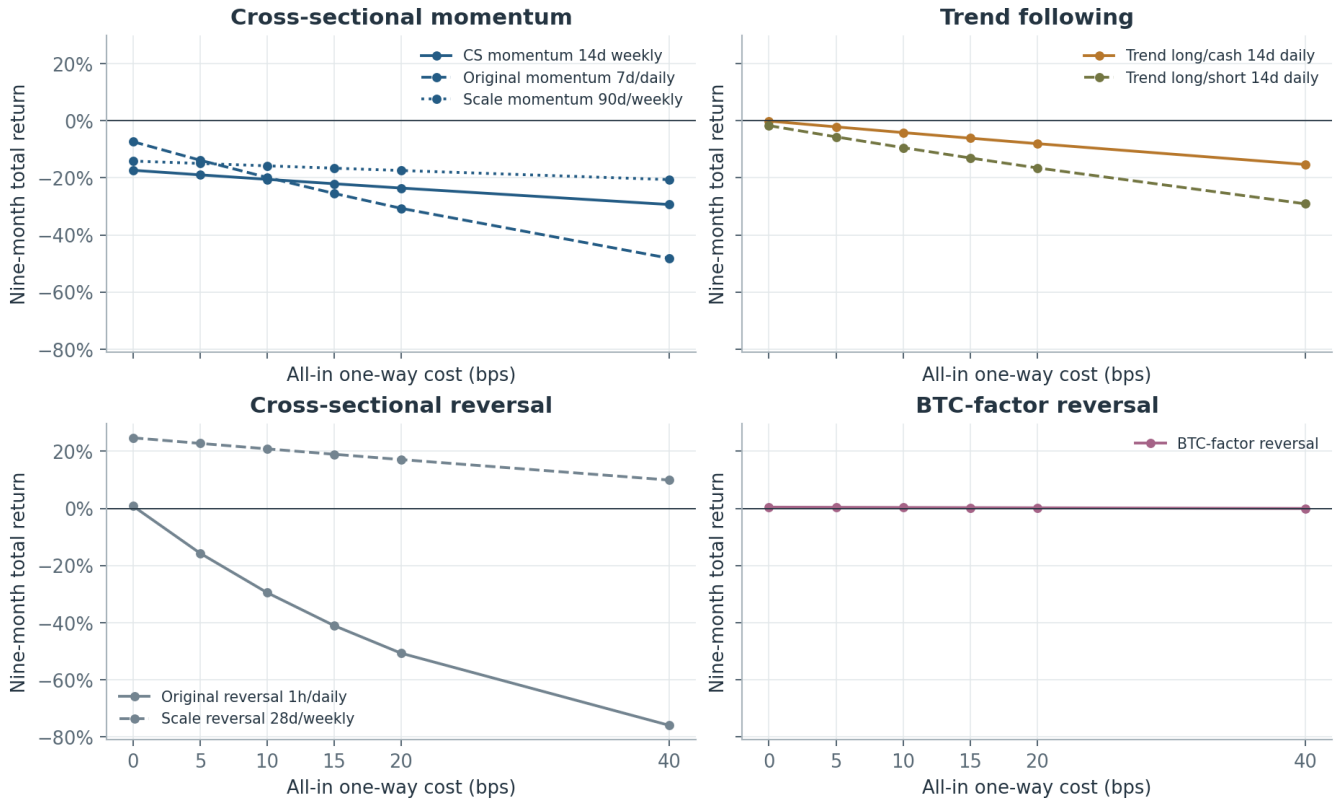


Frozen model	Gross return	Net return	Volatility	Sharpe	Max DD
CS momentum 14d weekly	-17.37%	-23.59%	15.68%	-2.21	25.11%
BTC-factor reversal	0.46%	0.23%	0.98%	0.32	1.04%
Original momentum 7d/daily	-7.36%	-30.70%	16.78%	-2.84	32.27%
Original reversal 1h/daily	0.82%	-50.68%	14.63%	-6.38	52.31%
Scale momentum 90d/weekly	-14.14%	-17.45%	17.76%	-1.35	19.82%
Scale reversal 28d/weekly	24.73%	17.11%	14.53%	1.53	8.87%
Trend long/cash 14d daily	-0.14%	-8.04%	31.82%	-0.20	33.66%
Trend long/short 14d daily	-1.75%	-16.53%	49.33%	-0.25	50.32%

January 1-September 30, 2026, primary 20 bps / lag one hour. Total returns are compounded over 273 days; volatility and Sharpe use daily returns and 365-day annualization. Drawdown is the maximum observed-endpoint loss magnitude. Curves share the same equity axis and begin at one. Line style distinguishes controls within the same family. N/A is undefined, never zero.

These eight paths are evaluations of previously frozen choices, not an invitation to select whichever worked best in 2026. A favorable historical outcome still needs prospective confirmation and realistic short/execution feasibility.

07 / Costs, execution timing, and exposure



Cost scenarios reuse identical signals and positions. Zero bps isolates gross behavior; 5 bps is below the course-assumed 7 bps commission alone. Cost/timing scenarios are model stresses, not confidence intervals. No 2026 selection or parameter rescue is performed.

Frozen model	Net lag 1	Net lag 2	Mean gross	Mean net	BTC beta
CS momentum 14d weekly	-23.59%	-26.00%	1.00	0.00	-0.02
BTC-factor reversal	0.23%	0.23%	0.03	-0.00	-0.00
Original momentum 7d/daily	-30.70%	-31.61%	1.00	0.00	-0.05
Original reversal 1h/daily	-50.68%	-51.25%	1.00	-0.00	0.03
Scale momentum 90d/weekly	-17.45%	-18.36%	1.00	0.00	-0.08
Scale reversal 28d/weekly	17.11%	17.42%	1.00	0.00	0.01
Trend long/cash 14d daily	-8.04%	-10.83%	0.50	0.50	0.39
Trend long/short 14d daily	-16.53%	-21.65%	1.00	0.00	-0.35

Selected long/cash mean uninvested allocation: 49.91%. This cash share is not redistributed to other coins.

Long/cash retains inactive cash and may have variable directional exposure. Long/short trend following is not necessarily dollar-neutral. Factor reversal allocates each spread including its hedge, but aggregated exposure can be below one; estimated factor hedging does not guarantee realized neutrality. The notebook records cash participation, CAGR, annual arithmetic return, turnover, and alpha with HAC intervals.

08 / Annualized metrics and factor exposure

Frozen model	CAGR	Arithmetic annual	Turnover	Annual alpha	Alpha 95% HAC interval
CS momentum 14d weekly	-30.22%	-34.73%	39.1	-34.64%	-70.38% to 1.10%
BTC-factor reversal	0.31%	0.31%	1.1	0.31%	-1.57% to 2.19%
Original momentum 7d/daily	-38.75%	-47.59%	145.0	-47.38%	-86.17% to -8.59%
Original reversal 1h/daily	-61.14%	-93.31%	357.0	-93.43%	-125.17% to -61.68%
Scale momentum 90d/weekly	-22.61%	-24.04%	19.6	-23.72%	-63.00% to 15.56%
Scale reversal 28d/weekly	23.51%	22.16%	31.5	22.13%	-12.36% to 56.63%
Trend long/cash 14d daily	-10.61%	-6.22%	41.2	-7.89%	-66.34% to 50.55%
Trend long/short 14d daily	-21.46%	-12.15%	81.5	-10.64%	-116.42% to 95.15%

These quantities have different meanings

CAGR extrapolates the nine-month compounded return to an annualized rate; it is not a realized twelve-month return. Arithmetic annual return is daily mean x 365. Turnover is total traded notional per unit of target capital over the actual nine-month interval, including entry and liquidation, not annualized or halved.

Alpha is the intercept from daily OLS net portfolio return against same-day BTC return, multiplied by 365. Its 95% interval uses HAC standard errors with seven daily lags and is annualized consistently. Beta is the regression slope, not a target hedge weight. Dollar neutrality does not establish zero realized beta; long/cash and own-trend long/short can carry market exposure.

Undefined quantities

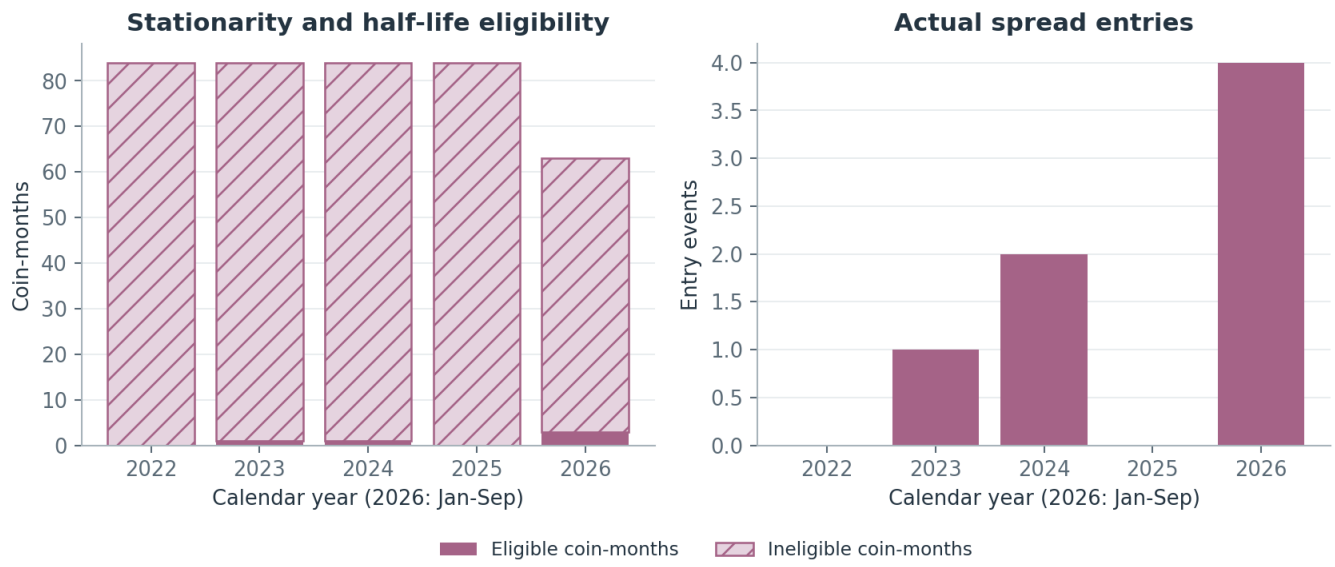
N/A denotes an undefined or unavailable estimate. A truly all-cash path has zero return, volatility, and turnover, but its Sharpe is undefined rather than zero. Constant portfolio returns can also leave some fitted uncertainty degenerate. No undefined metric is silently replaced by a favorable score.

Model identities

Canonical identifier	Frozen role
cross_momentum_h336_r168_b000	champion
trend_long_cash_h336_r024_b000	champion
trend_long_short_h336_r024_b000	champion
factor_residual_fixed	fixed model
reference_original_momentum	original reference
reference_original_reversal	original reference
reference_scale_momentum	time scale reference
reference_scale_reversal	time scale reference

The original momentum/reversal and time-scale controls retain their original selected horizons/cadences. Reference calculations here use the same holdout calendar and cost convention as champions; they are not new discoveries or fresh 2026 selections.

09 / BTC-factor residual reversal



Formation and eligibility

At each month start, fit using 90 daily observations ending 24 hours earlier. Estimate return beta on the earliest 30 observations; calibrate the fixed-beta log-price spread on the subsequent 60 endpoints. Require beta in [0.1, 5], positive spread variance, Holm-adjusted ADF significance at 5% across seven coins, and AR(1) half-life between one and 20 days. Ineligible coin-months remain cash; parameters are frozen within the trading month.

Trading state machine

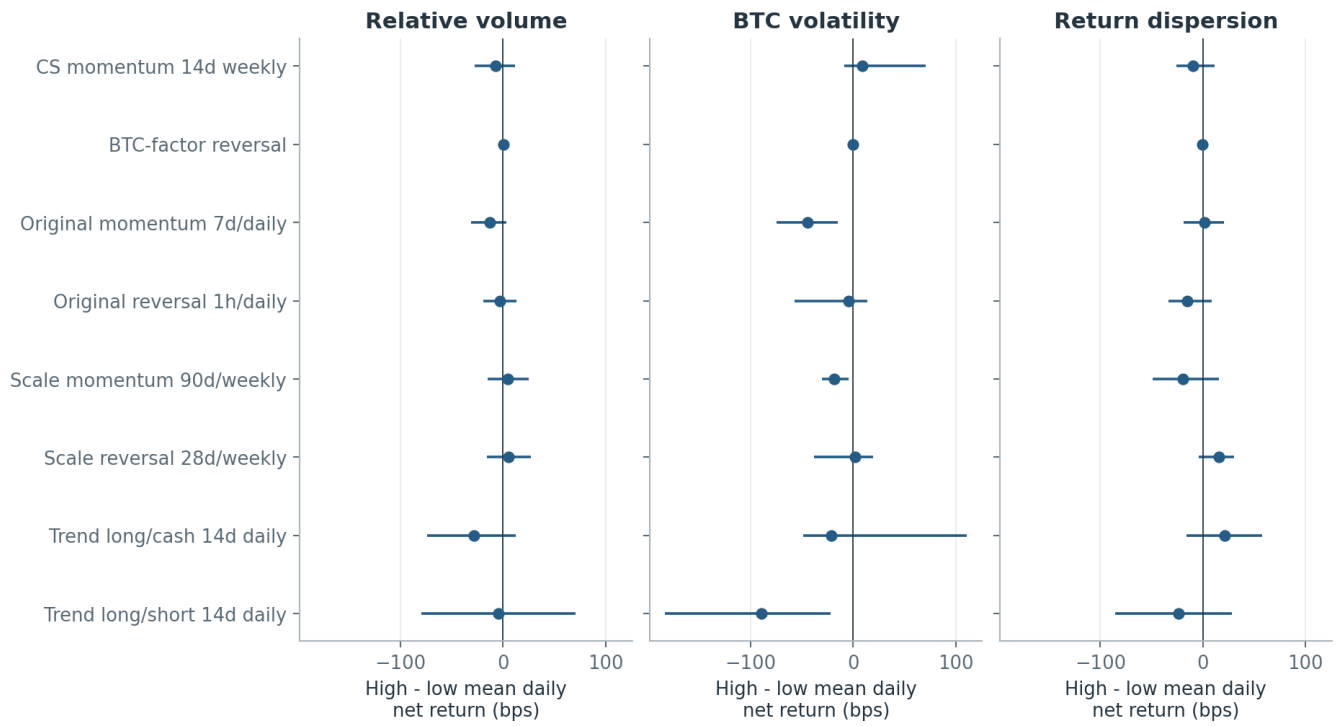
Enter toward the fitted mean only when 2 <= |z| < 4. Exit after crossing the signed 0.5 boundary, reaching an adverse signed 4 stop, or holding 20 calendar days. No same-day reentry. Force cash and refit on the first day of every month; allow entries from the following day. Each non-BTC coin spread receives at most 1/7 capital including its beta-weighted BTC hedge. Aggregate all BTC legs before cost accounting.

How to interpret inactivity

A valid all-cash or sparse result means this predeclared model did not find enough eligible tradable spread deviations. It is not missing data and should not prompt silent relaxation of stationarity, entry, or half-life thresholds. An ADF finding over 60 observations does not establish stable future cointegration.

Holdout model log: 3 eligible coin-months out of 63 scheduled coin-months. Every eligibility result and entry/exit reason is saved in factor_models.csv and factor_events.csv. Raw logs retain seven unused October 1 terminal refits; these are excluded from Jan-Sep chart/counts and no earned holdout return uses their new models.

10 / Regimes are diagnostics, not trade filters



Regime definitions

Relative quote volume, trailing 30-day BTC volatility, and cross-coin return dispersion are classified high/low using medians fitted only to 2022-2023. Labels are available at decision time and relate to the following day's return and turnover, not same-day outcomes. Unknown regime observations are preserved rather than assigned to low activity.

Reading the estimates

Each point is high-regime minus low-regime mean daily net return at 20 bps; horizontal bounds are nominal block-bootstrap 95% intervals, not Sharpe ratios or cumulative regime equity. Bootstrap uses the declared 28-day blocks and seed; Holm correction addresses the regime contrasts within each period. Sample counts and corrected p-values are retained in the notebook and regime_summary.csv.

The contrasts describe association in this fixed universe. Volume, volatility, and dispersion can move together; these cuts do not identify information-driven trading, establish causal effects, or justify market-timing gates. Sparse high/low subsets produce wide uncertainty. Missing intervals are not evidence of a zero contrast.

Holdout multiplicity check: 0 of 24 high-minus-low contrasts pass Holm 5%; nominal uncorrected intervals should not be promoted to validated regime effects.

No annualized Sharpe is computed by concatenating noncontiguous high- or low-regime days. All models trade according to their own frozen rule irrespective of these labels.

11 / What the research does and does not establish

Scope of the conclusion

The extension answers five concrete research questions with reproducible evidence. It does not search unrestricted coin pairs, tune a volatility gate, estimate an implementable borrowing/funding system, or claim risk-free arbitrage. Momentum/reversal are signal definitions; statistical arbitrage is the course umbrella label. Dollar neutrality and beta neutrality are distinct.

Material limitations

The eight surviving instruments are fixed after their survival was known. The 2022-2025 period has already been inspected in earlier research. Configuration screening can capitalize on noise; 2024 winner metrics are not unbiased forecasts. A single nine-month historical holdout is sensitive to market path and cannot establish persistence. Past-data monthly residual refits are permissible but their estimates can be unstable.

Funding, borrow availability/fees, margin/collateral, taxes, measured spread and impact, exchange outages, order-book depth, guaranteed fills, and ordinary target-weight drift are excluded. The close-based one-hour lag is an idealized execution model. The 5/10/15 bps scenarios should not be treated as verified real-world fees. No live orders, public publication, or scheduled monitoring occurred.

Reproducibility and preservation

The source package, configuration, all tested configurations, model logs, raw-archive checksums, data-quality receipts, freeze receipt, and output hashes accompany this report. Reporting uses frozen outputs and is separate from strategy evaluation. Original report, notebook, results, time-scale scan, and ZIP are preserved; original hashes are recorded in baseline_preservation.json.

Sources and attribution

Binance public archive documentation
https://github.com/binance/binance-public-data

Binance dataset terms and CC BY-NC-SA 4.0
https://github.com/binance/binance-public-data/blob/master/TERMS_AND_CONDITIONS.md

Binance March 24, 2023 halt account
https://www.binance.com/en/blog/from-our-ceo/6789340645608890113

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Run commands

From the separate extension project, use the installed virtual environment. The runner stages enforce history -> screening -> freeze -> holdout -> report. For artifact regeneration only: python -m crypto_stat_arb_extensions report. This does not fetch fresh data or change selections.